

Statistical Computing

Monte Carlo Methods in Inference

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Introduction

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Introduction

Monte Carlo methods: computational tools based on random sampling.

- Estimate sampling distributions, MSE, percentiles, etc.
- Assess coverage probability, Type I error rates, power of tests
- Compare statistical procedures
- Use repeated sampling from a given model (*parametric bootstrap*)

Monte Carlo Methods for Estimation

Setup

Suppose $X_1, \dots, X_n$ is a random sample from the distribution of X and let

$$T = T(X_1, \dots, X_n) \quad (1)$$

be an estimator of a parameter θ . Let $\mathbf{x} = (x_1, \dots, x_n)^\top \in \mathbb{R}^n$, and let $\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \dots$ denote a sequence of independent random samples generated from the distribution of X .

Random variates from the sampling distribution of T can be generated by repeatedly drawing $\mathbf{x}^{(j)}$ and computing

$$T^{(j)} = T(\mathbf{x}^{(j)}).$$

Monte Carlo estimate of an expectation

Let $\theta = \mathbb{E}[g(X_1, \dots, X_n)]$. Based on m independent replicates,

$$\hat{\theta} = \frac{1}{m} \sum_{j=1}^m g(\mathbf{x}^{(j)}). \quad (2)$$

Example. If $X_1, X_2 \stackrel{\text{iid}}{\sim} N(0, 1)$, then $\theta = \mathbb{E} |X_1 - X_2|$ can be estimated by averaging $|x_1^{(j)} - x_2^{(j)}|$ over m independent samples.

Example output (as shown in the text)

One run gives $\hat{\theta} \approx 1.128402$.

The exact values (by integration) are

$$\mathbb{E}|X_1 - X_2| = \frac{2}{\sqrt{\pi}} = 1.128379, \quad \text{Var}(|X_1 - X_2|) = 2 - \frac{4}{\pi}, \quad (3)$$

$$se(\hat{\theta}) = \sqrt{\frac{2 - 4/\pi}{m}} = 0.02695850 \quad (m = 1000). \quad (4)$$

Estimating the standard error of the mean

For a sample $x_1, \dots, x_m$ with mean $\bar{x}$, the plug-in estimate of the variance is

$$\widehat{\text{Var}}(x) = \frac{1}{m} \sum_{i=1}^m (x_i - \bar{x})^2, \quad (5)$$

and the corresponding estimate of the standard error of $\bar{x}$ is

$$\widehat{\text{se}}(\bar{x}) = \frac{1}{\sqrt{m}} \left[\frac{1}{m} \sum_{i=1}^m (x_i - \bar{x})^2 \right]^{1/2} = \frac{1}{m} \left[\sum_{i=1}^m (x_i - \bar{x})^2 \right]^{1/2}. \quad (6)$$

Using the unbiased variance estimator,

$$\widehat{\text{se}}(\bar{x}) = \frac{1}{\sqrt{m}} \left[\frac{1}{m-1} \sum_{i=1}^m (x_i - \bar{x})^2 \right]^{1/2}. \quad (7)$$

Monte Carlo standard error

Let

$$Y_j = g(\mathbf{x}^{(j)}), \quad j = 1, \dots, m.$$

Then the Monte Carlo estimator can be written as

$$\hat{\theta} = \bar{Y} = \frac{1}{m} \sum_{j=1}^m Y_j.$$

Its estimated standard error is

$$\widehat{se}(\hat{\theta}) = \frac{1}{m} \left[\sum_{j=1}^m (Y_j - \bar{Y})^2 \right]^{1/2}.$$

The mean squared error (MSE) of an estimator $\hat{\theta}$ for θ is

$$\text{MSE}(\hat{\theta}) = \mathbb{E}[(\hat{\theta} - \theta)^2]. \quad (8)$$

If m (pseudo) random samples $\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(m)}$ are generated and $\hat{\theta}^{(j)} = \hat{\theta}(\mathbf{x}^{(j)})$, then the Monte Carlo estimate is

$$\widehat{\text{MSE}}(\hat{\theta}) = \frac{1}{m} \sum_{j=1}^m (\hat{\theta}^{(j)} - \theta)^2. \quad (9)$$

Example: MSE of a trimmed mean

Level- k trimmed sample mean

Let $X_{(1)} \leq \dots \leq X_{(n)}$ be the ordered sample. The level- k trimmed mean is

$$X_{[-k]} = \frac{1}{n - 2k} \sum_{i=k+1}^{n-k} X_{(i)}. \quad (10)$$

Suppose $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(0, 1)$ and let $T = X_{[-1]}$ be the level-1 trimmed mean.

In this example, the target parameter is the center of the distribution:

$$\theta = \mathbb{E}[X] = \mathbb{E}[T] = 0.$$

Hence

$$\text{MSE}(T) = \mathbb{E}[(T - \theta)^2] = \mathbb{E}[T^2].$$

Example output (as shown in the text)

For the level-1 trimmed mean with $n = 20$ and $m = 1000$:

$$\widehat{\text{MSE}}(T) \approx 0.05176, \quad \widehat{\text{se}} \approx 0.00719.$$

For comparison, for the sample mean $\bar{X}$ under standard normal sampling, $\text{MSE}(\bar{X}) = \text{Var}(\bar{X}) = 1/n = 1/20 = 0.05$.

Estimating a confidence level

If (U, V) is a confidence interval for θ , then U and V are statistics whose distributions depend on the sampled population F_X . The true confidence level is the coverage probability

$$\mathbb{P}(\theta \in (U, V)).$$

- Evaluating this probability is a cdf / integration problem for the sampling distribution of the interval endpoints.
- Under the assumed model this may be tractable, but under non-normal sampling the true coverage can be unknown or intractable.
- Monte Carlo avoids direct calculation: repeatedly generate samples, construct a confidence interval C from each sample, and average the indicators $I(\theta \in C)$.

Monte Carlo experiment: empirical confidence level

Procedure

Suppose $X \sim F_X$ and θ is the target parameter. For $j = 1, \dots, m$:

1. Generate a random sample $X_1^{(j)}, \dots, X_n^{(j)}$.
2. Compute a confidence interval C_j from the j th sample.
3. Set $y_j = I(\theta \in C_j)$.

Then the empirical confidence level is $\bar{y} = \frac{1}{m} \sum_{j=1}^m y_j$.

Since $\bar{y}$ is a sample proportion, $\widehat{se}(\bar{y}) = \sqrt{\bar{y}(1 - \bar{y})/m}$.

Classical result under normal sampling

If $X_1, \dots, X_n$ is a random sample from $\text{Normal}(\mu, \sigma^2)$ ($n \geq 2$) and S^2 is the sample variance, then

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1). \quad (11)$$

A one-sided $100(1 - \alpha)\%$ upper confidence interval for σ^2 is

$$\left(0, \frac{(n-1)S^2}{\chi_{\alpha}^2(n-1)} \right), \quad (12)$$

where $\chi_{\alpha}^2(n-1)$ is the α -quantile of $\chi^2(n-1)$.

$n = 20$, $\alpha = 0.05$, and the sampled population is $N(0, \sigma^2 = 4)$.

- Several runs give upper confidence limits such as 6.628, 7.348, and 9.621; all contain $\sigma^2 = 4$.
- If the sampling and interval construction are repeated many times, approximately 95% of the intervals should contain $\sigma^2 = 4$.

Monte Carlo Methods for Hypothesis Tests

Suppose we wish to test a hypothesis concerning a parameter θ in a parameter space Θ :

$$H_0 : \theta \in \Theta_0 \quad \text{vs} \quad H_1 : \theta \in \Theta_1, \quad (13)$$

where Θ_0 and Θ_1 partition Θ .

Two types of error:

- Type I error: reject H_0 when H_0 is true.
- Type II error: do not reject H_0 when H_0 is false.

- **Significance level** α : upper bound on $\mathbb{P}(\text{Type I error})$

$$\alpha = \sup_{\theta \in \Theta_0} \pi(\theta), \quad \text{where } \pi(\theta) = \mathbb{P}(\text{reject } H_0).$$

- If T is the test statistic and T^* is its observed value, then T^* is **significant** if the test decision based on T^* is to reject H_0 .
- **p-value**: smallest α at which the observed T^* would be significant.

Empirical Type I error rate

Monte Carlo experiment

For each replicate $j = 1, \dots, m$:

1. Generate sample $\mathbf{x}^{(j)}$ from the null distribution.
2. Compute test statistic T_j .
3. Record $I_j = 1$ if H_0 rejected, else $I_j = 0$.

Empirical Type I error rate: $\hat{p} = \frac{1}{m} \sum_{j=1}^m I_j$.

Standard error: $\widehat{se}(\hat{p}) = \sqrt{\hat{p}(1 - \hat{p})/m}$.

Example: Empirical Type I error rate for a one-sided t -test

Suppose $X_1, \dots, X_{20}$ is a random sample from a $N(\mu, \sigma^2)$ distribution. Test

$$H_0 : \mu = 500 \quad \text{vs} \quad H_1 : \mu > 500 \quad \text{at } \alpha = 0.05. \quad (14)$$

Under the null hypothesis, $T^* = \frac{\bar{X} - 500}{S/\sqrt{20}} \sim t(19)$, where $t(19)$ denotes the Student t distribution with 19 degrees of freedom. Large values of T^* support the alternative hypothesis.

Goal: Use a Monte Carlo method to compute an empirical probability of Type I error when $\sigma = 100$, and check that it is approximately equal to $\alpha = 0.05$.

Example output (as shown in the text)

The empirical Type I error rate and its standard error:

$$\hat{p} \approx 0.0506, \quad \widehat{se}(\hat{p}) \approx 0.00219.$$

Theoretically, the probability of rejecting the null hypothesis when $\mu = 500$ is exactly $\alpha = 0.05$ in this example.

Example: Skewness test of normality

The skewness γ_1 of a random variable X is defined by

$$\gamma_1 = \frac{\mathbb{E}[(X - \mu_X)^3]}{\sigma_X^3}, \quad \mu_X = \mathbb{E}[X], \quad \sigma_X^2 = \text{Var}(X). \quad (15)$$

- A distribution is symmetric if $\gamma_1 = 0$.
- Positively skewed if $\gamma_1 > 0$, negatively skewed if $\gamma_1 < 0$.

The sample coefficient of skewness $\hat{\gamma}_1$ is

$$\hat{\gamma}_1 = \frac{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^3}{\left(\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2\right)^{3/2}}. \quad (16)$$

$$H_0 : \gamma_1 = 0 \quad \text{vs} \quad H_1 : \gamma_1 \neq 0.$$

If X is normal, then $\hat{\gamma}_1$ is asymptotically normal with mean 0 and variance $6/n$.

Skewness test: critical value

The test rejects for large $|\hat{\gamma}_1|$. Under the normal limit distribution,

$$cv_n = z_{0.975} \sqrt{\frac{6}{n}}, \quad (17)$$

and H_0 is rejected if $|\hat{\gamma}_1| > cv_n$.

We assess whether this asymptotic test achieves the nominal level $\alpha = 0.05$ for $n = 10, 20, 30, 50, 100, 500$.

- Under normality, the asymptotic distribution does not depend on μ or σ^2 .
- Hence it is enough to generate samples from $N(0, 1)$ and compute the rejection proportion.

Example output (as shown in the text)

n	10	20	30	50	100	500
$\hat{p}_{\text{reject}}$	0.0129	0.0272	0.0339	0.0415	0.0464	0.0539

With $m = 10000$ replicates, the Monte Carlo standard error is approximately

$$\sqrt{\frac{0.05 \cdot 0.95}{m}} = 0.0022.$$

- The asymptotic test is too conservative for small and moderate sample sizes.
- The approximation is still questionable even at $n = 500$.

- **Type II error:** fail to reject H_0 when H_1 is true.
- **Power** $\pi(\theta)$: probability of rejecting H_0 when true parameter is θ .
- For alternative θ_1 : $\mathbb{P}(\text{Type II error}) = 1 - \pi(\theta_1)$.
- Type I error controlled by α ; high power $\Leftrightarrow$ low Type II error.

Monte Carlo experiment to estimate power

Against a fixed alternative $\theta_1 \notin \Theta_0$

1. Select a particular parameter value $\theta_1 \notin \Theta_0$.
2. For each replicate $j = 1, \dots, m$:
 - (a) Generate the j th random sample under the alternative $\theta = \theta_1$.
 - (b) Compute the test statistic T_j .
 - (c) Record the test decision: set $I_j = 1$ if H_0 is rejected at significance level α , otherwise set $I_j = 0$.
3. Compute the proportion of significant tests

$$\hat{\pi}(\theta_1) = \frac{1}{m} \sum_{j=1}^m I_j. \quad (18)$$

- To compare power across alternatives, repeat over a grid of values θ .

Example: Empirical power curve for a one-sided t -test

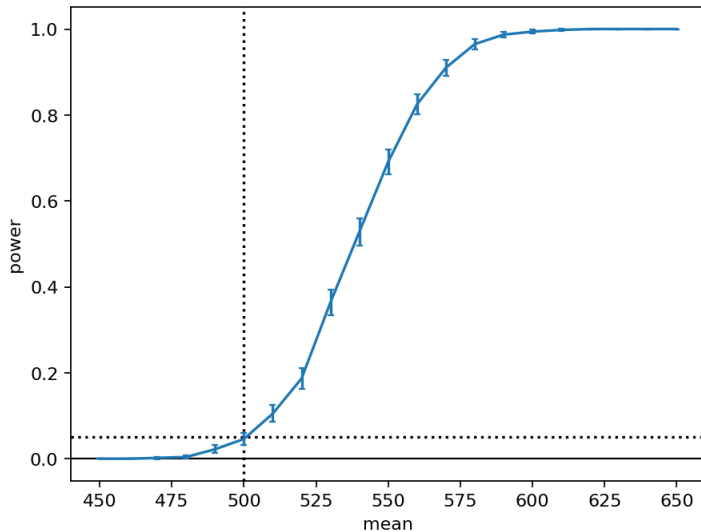
One-sided t -test with $n = 20$ and $\alpha = 0.05$:

$$H_0 : \mu = 500 \quad \text{vs} \quad H_1 : \mu > 500, \quad \text{with } \sigma = 100. \quad (19)$$

- Compute $\hat{\pi}(\mu)$ on a grid of true means $\mu = 450, 460, \dots, 650$.
- Each point on the curve is one Monte Carlo experiment; for $\mu > 500$, $\hat{\pi}(\mu)$ is the empirical power.
- For $\mu < 500$, the same plot shows how small the rejection probability is to the left of the boundary $\mu_0 = 500$.

The Monte Carlo standard error can be estimated by $\widehat{se}(\hat{\pi}(\mu)) = \sqrt{\hat{\pi}(\mu)(1 - \hat{\pi}(\mu))/m}$.

Empirical power curve: figure



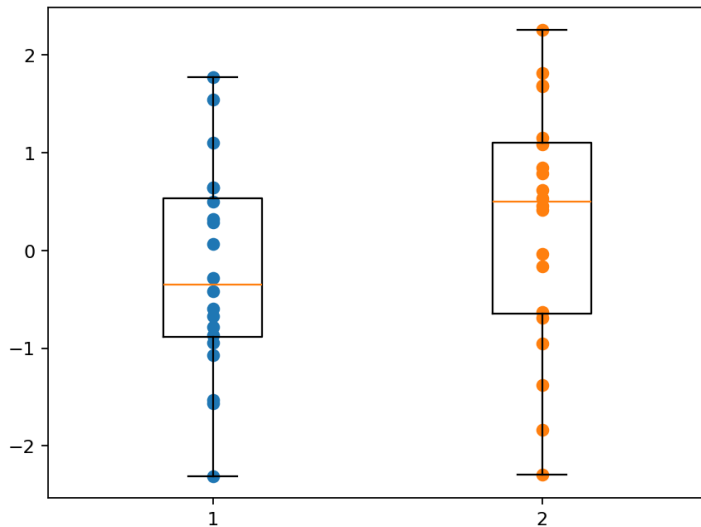
Application: “Count Five” Test for Equal Variance

Application: “Count Five” test for equal variance

A simple two-sample test for equal variance due to McGrath and Yeh:

- An observation is *extreme* if it lies outside the range of the other sample.
- The Count Five statistic is the maximum number of extreme points in the two samples.
- The test is intended for similarly distributed populations (same shape up to location and scale), with equal means and equal sample sizes.

Count Five statistic: figure



Count Five statistic: numerical example

Ranges and extreme points:

$$\text{range}(x_1) : (-2.783, 1.729), \quad \text{range}(x_2) : (-1.599, 3.710).$$

$$x_1[i] : -2.783, \quad x_2[j] : 2.036, 1.810, 3.710.$$

The Count Five statistic is the maximum number of extreme points, $\max(1, 3) = 3$, so the test does not reject equal variance in this example.

Estimating the sampling distribution of the extreme-count statistic

Consider two independent samples from the same normal distribution with $n_1 = n_2 = 20$.

- Under H_0 , estimate the sampling distribution of the maximum number of extreme points.
- Find the 0.80, 0.90, and 0.95 quantiles.
- Compute the statistic by centering each sample and counting extremes relative to the range of the other centered sample.

Extreme-count distribution under H_0

Empirical cdf:

stat	1	2	3	4	5	6	7	8	9	10	11
$\hat{F}$	0.149	0.512	0.748	0.871	0.945	0.974	0.986	0.990	0.996	0.999	1.000

Quantiles: 80% : 4, 90% : 5, 95% : 6.

If $\alpha = 0.05$ is desired, the empirical cdf suggests using the practical rule

reject if the extreme-count statistic > 5 .

So the quantile function alone is not always the best way to choose a critical value.

Count Five test with centering

- When centered by population means, McGrath and Yeh show that the significance level is at most 0.0625.
- In practice, the population means are unknown, so each sample is centered by its sample mean.

For normal samples with $n_1 = n_2 = 20$, the empirical Type I error rate is

$$\hat{\alpha} \approx 0.0565 \quad (\hat{se} \approx 0.0022),$$

which is close to the population-mean-centered benchmark above.

Unequal sample sizes

When sample sizes are unequal, Type I error is **not** controlled:

- $n_1 = 20, n_2 = 30$: $\hat{\alpha} \approx 0.1064$
- $n_1 = 20, n_2 = 50$: $\hat{\alpha} \approx 0.2934$

⇒ The unadjusted Count Five criterion should not be used with unequal sample sizes.